

At Standard Training Budgets, SAE Features Match Random Baseline: Evidence for Underconstrained Sparse Reconstruction

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Abstract

Sparse Autoencoders (SAEs) have emerged as a leading tool for mechanistic interpretability, yet the reproducibility of their learned features across training runs remains poorly characterized. We present a multi-seed stability analysis with random-baseline controls, training dynamics through 2,000 epochs, and cross-scale validation on Pythia-70M/160M/410M. At standard training budgets (20 epochs), trained SAE features are practically indistinguishable from randomly initialized SAEs (PWMCC 0.300 vs 0.299, TOST equivalence testing indicates practical equivalence at $\varepsilon = 0.005$, $p < 0.001$; 20 seeds, 190 pairwise comparisons), despite achieving strong reconstruction (explained variance > 0.91). However, extended training reveals monotonic convergence: TopK SAEs asymptote at 0.732 PWMCC (2,000 epochs, $2.44\times$ random) while ReLU plateaus at 0.666. This dissociation — excellent reconstruction with training-budget-dependent stability — reveals that the sparse reconstruction objective is underconstrained, but the optimization landscape does contain a partially shared solution region that sufficient training can approach. Optimal-transport alignment achieves 0.881 PWMCC on mod-113 with only 5.9% reconstruction cost, exceeding Song et al.’s 0.80 consistency target. The method generalizes to Pythia-410M (+37% PWMCC, 1.6% reconstruction cost), though reaching the 0.80 threshold at LLM scale remains open. These findings challenge interpretability claims built on undertrained SAE features and provide concrete guidance: measure effective rank, train sufficiently, and consider alignment-aware objectives.

Keywords: Sparse Autoencoders, Mechanistic Interpretability, Feature Stability, Reproducibility

1. Introduction

Sparse Autoencoders promise to decompose neural networks into human-interpretable features, with recent applications scaling to frontier language models (Templeton et al., 2024; Gao et al., 2024). The fundamental premise is that SAEs can uncover “the true underlying features used by a model” (Elhage et al., 2022). Yet a critical question threatens this enterprise: do SAE features generalize across training runs, or are they artifacts of random initialization?

If features vary arbitrarily with random seeds, interpretations based on single SAE instances may be meaningless — the features discovered could simply be one of many equally valid decompositions, with no claim to uniqueness. This concern was recently validated by Paulo & Belrose (2025), who found that only 30% of features are shared across independently trained SAEs on large language models, and by Song et al. (2025), who argued that feature consistency should be elevated to a primary evaluation criterion. Independent work has shown that SAEs trained on randomly initialized (untrained) transformers produce similar auto-interpretability scores to those from trained models (Heap et al., 2025), and that SAE features do not constitute canonical units of analysis (Leask et al., 2025). These findings collectively question whether standard SAE features reflect genuine model structure.

1.1 Research Questions

We address four fundamental questions:

1. **Do SAE features generalize across training runs, or match random baseline?** We compare feature similarity (PWMCC) between trained SAEs against a critical control: randomly initialized, untrained SAEs.
2. **Are SAEs functionally successful despite representational instability?** We measure whether SAEs achieve good reconstruction even when features are unstable, testing whether the optimization landscape is underconstrained.
3. **Does this phenomenon generalize across model scales?** We validate on algorithmic transformers and Pythia-70M/160M/410M to test whether richer semantic structure changes the picture.
4. **Does feature stability improve with extended training?** We track PWMCC over 2,000 epochs to determine whether the random-baseline result reflects a fundamental property or an undertrained regime.

1.2 Key Contributions

Our work makes five primary contributions:

1. **Quantification of the random baseline phenomenon.** At standard training budgets (20 epochs), trained SAE feature similarity (PWMCC = 0.300) is practically indistinguishable from randomly initialized SAEs (PWMCC = 0.299, pairwise-comparison TOST, supported by a permutation test, indicates practical equivalence at $\epsilon = 0.005$ with 20 seeds and 190 pairwise comparisons), consistent across four architectures (TopK, ReLU, JumpReLU, BatchTopK).
2. **Training dynamics reveal convergence.** Extended training to 2,000 epochs shows monotonic PWMCC increase: TopK asymptotes at 0.732 (2.44x random) and ReLU at 0.666. The random-baseline result reflects insufficient training rather than a fundamental impossibility.
3. **Evidence for underconstrained reconstruction.** SAEs achieve excellent reconstruction (explained variance 0.92–0.98 vs ~ 0 for random), yet learn incompatible feature representations across seeds at standard budgets. This dissociation reveals that the sparse reconstruction task admits many equally good solutions, and standard training budgets are insufficient to navigate to a shared optimum.
4. **Cross-scale replication.** The overparameterization-stability pattern replicates on Pythia-70M, 160M, and 410M (Biderman et al., 2023): higher $d_{\text{sae}} / \text{eff_rank}$ is associated with lower stability at standard training budgets, though the strength of this relationship varies by model and layer.
5. **Stability-promoting methods work, with scale-dependent gains.** Optimal-transport alignment (following Song et al., 2025) achieves 0.881 PWMCC on mod-113 with only 5.9% reconstruction cost, exceeding Song et al.’s 0.80 target. On Pythia-410M, OT alignment reaches 0.653 PWMCC (+37% over standard, 6.1x random) with only 1.6% MSE cost, confirming the method generalizes to LLM-scale data but with diminished absolute gains in higher-dimensional settings.

2. Related Work

2.1 Sparse Autoencoders for Interpretability

SAEs have emerged as a leading tool for mechanistic interpretability (Cunningham et al., 2023; Bricken et al., 2023). Recent scaling efforts have applied SAEs to production models, including GPT-4 (Gao et al., 2024) and Claude 3 Sonnet (Templeton et al., 2024). Open-source SAE suites now cover Gemma 2 (Lieberum et al., 2024) and Llama 3.1 (Lieberum et al., 2024). Two major architectural variants dominate: ReLU SAEs with L1 sparsity penalty (Bricken et al., 2023) and TopK SAEs with hard sparsity (Gao et al., 2024), with recent innovations including JumpReLU (Rajamanoharan et al., 2024), BatchTopK (Bussmann et al., 2024), and Switch SAEs (Karvonen et al., 2024).

2.2 Feature Stability and Consistency

Paulo & Belrose (2025) trained 9 SAEs with different random seeds on Pythia 160M and Llama 3 8B, finding that only 30–42% of features are consistently recovered. Song et al. (2025) elevated this concern to a position paper, arguing that mechanistic interpretability should prioritize feature consistency, and demonstrated that 0.80 PWMCC is achievable with appropriate methods. Leask et al. (2025) showed through SAE stitching experiments that features are neither complete nor atomic, and Heap et al. (2025) demonstrated that SAEs trained on randomly initialized transformers produce similar interpretability scores to those from trained models – suggesting many features may reflect data statistics rather than learned computations.

Concurrent work. Several concurrent studies complement our findings. Korznikov et al. (2026) independently advocate random-baseline controls for SAE evaluation, focusing on benchmark-level sanity checks rather than our feature-level PWMCC analysis. Chanin & Garriga-Alonso (2026) introduce SynthSAEBench, demonstrating that reconstruction quality does not reliably predict feature quality – consistent with our functional-representational gap finding. Jedryszek & Crook (2026) show that weight regularization during SAE training improves feature stability, offering a complementary approach to the dictionary pinning and contrastive methods we evaluate.

2.3 Theoretical Foundations

Cui et al. (2026) characterized conditions under which SAE identifiability holds, identifying three key factors: extreme sparsity of ground truth, sparse SAE activation, and sufficient hidden dimensions. When these conditions are violated, the optimization landscape contains multiple equally good local minima. Their theory predicts low feature recoverability in dense, non-identifiable regimes; our PWMCC = 0.300 is consistent with this prediction. The SAE Bench evaluation framework (Karvonen et al., 2025) provides standardized metrics but does not include cross-seed stability as a primary criterion.

2.4 Grokking and Modular Arithmetic

We evaluate SAEs on a grokking transformer for modular arithmetic (Power et al., 2022; Nanda et al., 2023). This controlled setting offers 100% task accuracy for clean activation extraction. Nanda et al. (2023) showed that 1-layer transformers learn Fourier circuits ($R^2 = 93\text{--}98\%$); our 2-layer architecture does not rely on Fourier circuits ($R^2 = 2\%$), indicating that the instability we observe is not specific to the Fourier algorithm identified by Nanda et al.

3. Methods

3.1 Experimental Setup

Transformer training. We trained a 2-layer transformer ($d_{\text{model}} = 128$, $n_{\text{heads}} = 4$, $d_{\text{mlp}} = 512$) on modular addition (mod 113) to 100% accuracy via grokking (Power et al., 2022). Training used AdamW with learning rate $1e-3$ and weight decay 1.0 for 5000 epochs on 30% of possible pairs.

SAE architectures. We trained four SAE variants: - TopK: Hard sparsity with $k = 32$ active features (Gao et al., 2024) - ReLU: Soft sparsity with L1 penalty $\lambda = 1e-3$ (Bricken et al., 2023) - JumpReLU: Learned per-feature thresholds with straight-through estimator (Rajamanoharan et al., 2024) - BatchTopK: Batch-level top-k selection allowing variable per-sample sparsity (Bussmann et al., 2024)

All used expansion factor $8\times$ ($d_{\text{sae}} = 1024$ for $d_{\text{model}} = 128$), trained on layer 1 residual stream activations.

Multi-seed protocol. For each architecture, we trained 5 SAEs with seeds $\{42, 123, 456, 789, 1011\}$, ensuring identical data, hyperparameters, and training duration. Only random initialization differed.

3.2 Evaluation Metrics

Pairwise Maximum Cosine Correlation (PWMCC). Following Paulo & Belrose (2025) and Song et al. (2025), we compute cosine similarity matrix $S \in \mathbb{R}^{d_{\text{sae}} \times d_{\text{sae}}}$ between decoder columns, then average the maximum absolute similarity per feature across both directions. PWMCC ranges from 0 (no overlap) to 1 (perfect alignment).

Reconstruction metrics. Explained variance: $1 - \|x - \hat{x}\|^2 / \|x\|^2$. L0 sparsity: mean number of active features. Dead neuron percentage: features that never activate.

Statistical tests. Mann-Whitney U test (non-parametric) for group comparisons, with Cohen’s d effect sizes, and TOST (Two One-Sided Tests) equivalence testing to formally assess practical indistinguishability between trained and random PWMCC distributions. We set the equivalence margin at $\epsilon = 0.005$, corresponding to $\sim 5\times$ the observed standard deviation within each group (~ 0.001): a difference smaller than ϵ would be scientifically negligible at this measurement precision. Note that 5 seeds yield $C(5,2) = 10$ pairwise comparisons that are not fully independent, since each SAE participates in 4 pairs; the reported p-values should therefore be interpreted as suggestive rather than exact. The equivalence conclusion is robust to margin choice: because the 90% CI for the trained-vs-random difference ($[0.0004, 0.0020]$) falls within $\pm\epsilon$ for any $\epsilon \geq 0.002$, the TOST indicates equivalence at any margin above twice the observed difference. Our choice of $\epsilon = 0.005$ is conservative in this sense.

Effective rank. Following Roy & Vetterli (2007), we define effective rank as $\text{erank}(A) = \exp(-\sum_i \bar{\sigma}_i \log \bar{\sigma}_i)$, where $\bar{\sigma}_i = \sigma_i / \sum \sigma_i$ are the normalized singular values of the mean-centered activation matrix. This continuous measure equals the matrix rank when all nonzero singular values are equal and approaches 1 when a single component dominates.

3.3 Implementation

All experiments used PyTorch 2.0 with mixed precision training. SAEs were trained for 20 epochs with batch size 256 ($\sim 1,000$ gradient steps over 12,769 activation vectors), using AdamW ($\text{lr} = 3e-4$). Extended

training experiments (Section 4.6) explored up to 2,000 epochs (~100,000 gradient steps) on mod-113 and 200 epochs on Pythia models. All extended-training experiments used cached activations, amortizing the forward-pass cost. The mod-113 experiments ($d_{\text{model}} = 128$, 2,000 epochs, 5 seeds) complete in minutes on a single GPU. The Pythia-410M experiments ($d_{\text{model}} = 1024$, 200 epochs over 200K cached activations) require approximately 1–3 GPU-hours per seed on an NVIDIA B200, depending on d_{sae} . The 10–25 $\times$ training extension we recommend is thus practical for cached-activation SAE pipelines, scaling linearly with dictionary size and dataset. Experiment artifacts are logged as machine-readable JSON manifests with provenance metadata.

4. Results

4.1 Core Finding: Trained SAEs Match Random Baseline

At standard training budgets (20 epochs, ~1,000 gradient steps), SAE features match random baseline. We compared PWMCC between all pairs of 5 trained TopK SAEs against 5 randomly initialized, untrained SAEs. (Section 4.6 shows this result is training-budget-dependent.)

Comparison	PWMCC	Std	N pairs	Interpretation
Trained vs Trained	0.3001	0.0009	10	Cross-seed similarity
Random vs Random	0.2988	0.0011	10	Chance baseline
Difference	+0.0012			Practically zero

Statistical tests: Mann-Whitney $U = 82$, $p = 0.009$ (the small but statistically significant difference reflects the large sample precision, not practical importance). Pairwise-comparison TOST indicates practical equivalence at $\epsilon = 0.005$ ($p < 0.001$), with 90% CI for the difference: [0.0004, 0.0020]. These p-values treat the 10 pairwise comparisons as independent; since each SAE appears in 4 pairs, the effective sample size is smaller and the reported significance is suggestive (see §3.2). Nonetheless, the absolute difference of 0.001 is negligible, representing 0.4% of the PWMCC scale. At this training budget, feature representations show negligible stability above chance. A 20-seed robustness study ($C(20,2) = 190$ pairwise comparisons) confirms: TOST equivalence at $\epsilon = 0.005$ ($p < 0.001$), with a permutation test (10,000 permutations) yielding $p = 0.0001$, indicating that the difference, while statistically detectable, is practically negligible.

4.2 The Dissociation: Functional Success with Representational Instability

Despite matching random baseline in feature consistency, trained SAEs achieve dramatically better reconstruction:

Metric	Trained (TopK)	Trained (ReLU)	Random	Interpretation
Explained Variance	0.919	0.977	~0.0	Strong reconstruction
L0 Sparsity	32	~400	32	Controlled

SAEs are simultaneously functionally successful (explained variance > 0.91) and representationally unstable (PWMCC = random baseline). This complete decoupling demonstrates that reconstruction quality cannot

predict feature stability – a practitioner evaluating a single SAE would incorrectly conclude features are meaningful and reproducible.

4.3 Consistency Across Tested Architectures

All four tested architectures remain near the random baseline at standard training budgets:

Architecture	PWMCC	Over Random	L0
TopK	0.304 +/- 0.001	1.015x	32 (fixed)
BatchTopK	0.302 +/- 0.001	1.008x	32 (avg)
ReLU	0.300 +/- 0.001	1.002x	~328
JumpReLU	0.300 +/- 0.001	1.002x	~365

The instability is consistent across all four tested architectures. TopK and BatchTopK (which shares feature selection across the batch) show marginally higher PWMCC, but all four architectures remain within 1.5% of random baseline. This pattern replicates on Pythia-70M ($d_{\text{sae}} = 2048$, 3 seeds): all four architectures show PWMCC within 0.8% of random (TopK 1.008x, BatchTopK 1.002x, ReLU 1.001x, JumpReLU 1.001x). Unlike Paulo & Belrose (2025), who found TopK more unstable than ReLU on LLMs, we observe no practical difference on either algorithmic or Pythia-scale tasks at standard training budgets.

4.4 Effective Rank Study

A comprehensive d_{sae} sweep reveals a stability-reconstruction tradeoff governed by the activation effective rank (~ 80):

Regime	d_{sae}	k	PWMCC	Ratio to Random	Recon Loss
Under	16	4	0.513	2.87x	1.124
Under	32	8	0.454	2.22x	0.587
Under	48	12	0.406	1.87x	0.333
Matched	64	16	0.373	1.62x	0.203
Matched	80	20	0.355	1.51x	0.133
Matched	128	32	0.304	1.23x	0.052
Over	256	32	0.291	1.09x	0.026
Over	512	32	0.295	1.04x	0.028
Over	1024	32	0.304	1.02x	0.034

The tradeoff is clear: underparameterized SAEs ($d_{\text{sae}} < \text{eff_rank}$) achieve up to 2.87x random stability but sacrifice reconstruction, while overparameterized SAEs ($d_{\text{sae}} \gg \text{eff_rank}$) achieve near-perfect reconstruction but collapse to random stability.

4.5 Theoretical Grounding

Our empirical PWMCC = 0.300 is consistent with Cui et al.’s (2026) identifiability theory. Their analysis indicates that SAEs generally fail to recover ground-truth features unless: (1) the ground truth is extremely sparse, (2) SAE activation is sparse, and (3) hidden dimensions are sufficient.

Condition	Our Setup	Requirement	Status
Ground truth sparsity	Dense (eff_rank = 80/128 = 62.5%)	Extremely sparse (< 10%)	Violated
SAE activation sparsity	k = 32/1024 = 3.1%	k « ground truth sparsity	Marginal
Hidden dimensions	d_sae = 1024 » eff_rank = 80	d_sae >= ground truth dims	Met but harmful

Our dense activation subspace (62.5% of d_{model}) directly violates Condition 1. Theory therefore predicts low recoverability in our setting – our measured PWMCC of 0.300 is consistent with this prediction.

4.6 Training Dynamics

Extended training to 2,000 epochs on mod-113 reveals architecture-dependent asymptotic behavior:

Epoch	TopK PWMCC	ReLU PWMCC	TopK + Resampling
0 (random)	0.300	0.300	0.300
20	0.302	0.300	0.782
50	0.309	0.302	0.602
100	0.321	0.333	0.573
200	0.370	0.614	0.566
500	0.676	0.652	0.557
1000	0.723	0.659	0.570
2000	0.732	0.666	0.562

Three findings emerge. First, **TopK PWMCC asymptotes at ~0.73** (2.44x random), reaching 0.723 by epoch 1000 and gaining only +0.009 over the final 1000 epochs. The rapid climb from 0.676 (epoch 500) to 0.719 (epoch 750) followed by near-stagnation indicates a genuine stability ceiling below the 0.80 threshold. Second, **ReLU plateaus at ~0.66**, confirmed from epoch 300 through 2000 with no further improvement. Third, **dead neuron resampling hurts** at all timescales: it reaches 0.888 at epoch 10 but decays monotonically to 0.562 by epoch 2000, never recovering.

This resampling result is practically important because dead neuron resampling is standard practice in production SAE pipelines (Gao et al., 2024; Templeton et al., 2024). The mechanism is visible in the training dynamics: resampled features are initialized to align with underrepresented activation directions, producing artificially high cross-seed agreement at early epochs (0.888 at epoch 10). As training continues, these features adapt to their local loss landscape, re-diversifying across seeds. By epoch 50, resampled PWMCC (0.602) has already fallen below the non-resampled TopK trajectory, and the gap widens monotonically. The practical implication is that resampling optimizes for reconstruction coverage (reducing dead neurons) at the expense of cross-seed consistency. Practitioners who require feature stability should either avoid resampling or pair it with alignment-aware losses (Section 4.7.4).

Extended training generalizes to LLMs. We replicated the extended training experiment on Pythia-410M layer 21 and Pythia-160M layer 10 (200 epochs, 5 seeds each, batch_size=4096):

Model	Layer	Arch	Random BL	Ep200 PWMCC	Ratio	Trend
Pythia-410M	21	TopK	0.107	0.465	4.33x	Still climbing
Pythia-410M	21	ReLU	0.107	0.473	4.41x	Still climbing
Pythia-160M	10	TopK	0.124	0.259	2.09x	Plateaued ~ep100
Pythia-160M	10	ReLU	0.124	0.632	5.11x	Still climbing

The convergence pattern generalizes: both Pythia models show PWMCC climbing well above random with extended training. Notably, ReLU on Pythia-160M reaches 5.11x random at 200 epochs and continues climbing, while TopK on the same data plateaus early (2.09x). On Pythia-410M layer 21 (the most structured layer, `eff_rank` = 408.9), both architectures reach 4.3-4.4x random at 200 epochs with no sign of convergence, suggesting the asymptote may be substantially higher with further training.

The TopK-ReLU divergence on Pythia-160M layer 10 (TopK 2.09x vs ReLU 5.11x) contrasts with their convergence on Pythia-410M layer 21 (4.33x vs 4.41x). One possible explanation is that TopK’s hard sparsity constraint creates more local optima in lower-dimensional spaces (`d_model` = 768), causing earlier plateaus, while ReLU’s soft sparsity allows gradual redistribution of activation mass during extended training. However, we cannot rule out that TopK on Pythia-160M would resume climbing with substantially more training, and this architecture-scale interaction warrants further investigation.

The core claim must therefore be qualified: at standard training budgets (20 epochs), SAEs match random baseline. Extended training reveals monotonic convergence toward a partially shared optimum on both mod-113 and Pythia-scale models. However, on mod-113, TopK asymptotes at ~0.73 (below Song et al.’s 0.80 threshold), indicating that standard training alone may not reach full consistency.

4.7 Follow-Up Experiments

We conducted six additional experiments to probe the phenomenon’s boundaries.

4.7.1 Cross-Scale Validation: Pythia-70M, 160M, 410M To test whether the overparameterization pattern generalizes, we trained SAEs on Pythia-70M, 160M, and 410M (Biderman et al., 2023), extracting 200K activation vectors from multiple layers using wikitext-103 text.

Multilayer analysis on Pythia-70M. We swept layers {0, 2, 4, 5} of Pythia-70M (6 layers total, `d_model` = 512):

Layer	Eff. Rank	Best PWMCC/Random	At <code>d_sae</code> =4096
0	391.4	1.60x (<code>d_sae</code> =256)	1.07x
2	280.6	1.77x (<code>d_sae</code> =256)	1.09x
4	348.9	1.57x (<code>d_sae</code> =256)	1.10x
5	349.3	1.71x (<code>d_sae</code> =512)	1.11x

Layer 2 has the lowest effective rank (280.6, 54.8% of `d_model`) and shows the highest stability ratio at small `d_sae`. Layer 5 (pre-logit) shows notably higher stability at moderate `d_sae`. At `d_sae` / `eff_rank` > 5, all layers approach ~1.1x random.

Cross-scale replication. The overparameterization-stability tradeoff is visible across all three model scales, though the relationship varies in strength:

Model	Layer	d_model	Eff. Rank	Best Ratio	At d_sae/eff_rank ≈ 5
Pythia-70M	0	512	425.7	1.94x	1.19x
Pythia-160M	1	768	573.0	1.84x	1.51x
Pythia-160M	10	768	511.8	1.66x	1.31x
Pythia-410M	2	1024	757.2	2.09x	1.77x
Pythia-410M	21	1024	408.9	2.45x	1.54x

Two patterns emerge. First, **deeper layers with lower effective rank show higher stability**, consistent with more structured representations being easier to consistently decompose (Pythia-410M layer 21: eff_rank = 408.9, ratio = 2.45x vs layer 2: eff_rank = 757.2, ratio = 2.09x). Second, **absolute PWMCC remains low with short training** – even at the most favorable setting (Pythia-410M, layer 21, d_sae = 512), PWMCC is only 0.248 at 10 epochs. Higher d_sae / eff_rank is consistently associated with lower stability at this training budget across all 25 conditions tested, though the effect is model- and layer-dependent: stability ratios at d_sae / eff_rank ≈ 5 range from 1.19x to 1.77x random.

The d_sae / eff_rank threshold is a training budget artifact. To test whether overparameterized regimes are fundamentally unstable or merely undertrained, we swept d_sae in {512, 1024, 2048, 4096} on Pythia-410M layer 21 with 200 epochs of extended training (3 seeds each):

d_sae	d_sae / eff_rank	10-epoch ratio	200-epoch PWMCC	200-epoch ratio
512	1.3	2.45x	0.479	4.73x
1024	2.5	1.54x	0.458	4.27x
2048	5.0	$\sim 1.1x$	0.523	4.63x
4096	10.0	$\sim 1.0x$	0.537	4.54x

All d_sae values converge to 4.3-4.7x random with extended training, regardless of the d_sae / eff_rank ratio. The d_sae = 4096 condition (ratio 10.0), which showed near-random stability at 10 epochs, reaches 4.54x random at 200 epochs – indistinguishable from the d_sae = 512 condition. This demonstrates that **the threshold is an artifact of insufficient training, not a fundamental property of the optimization landscape**. In this sweep, the short-budget d_sae / eff_rank penalty disappears with extended training.

Reconciliation with Paulo & Belrose (2025). Our results and Paulo & Belrose’s findings on Pythia-160M are consistent rather than contradictory. We now test the same model family (Pythia-160M) and find stability ratios of 1.84x at d_sae = 512 with short training, consistent with their observation that a minority of features are shared. The qualitative pattern – stability decreasing with overparameterization – holds at standard training budgets but disappears with extended training.

4.7.2 1-Layer Ground Truth Comparison We compared SAE stability on 1-layer (eff_rank = 33.5) vs 2-layer (eff_rank = 80.5) transformers. Both show qualitatively similar behavior: stability peaks near d_sae = 2x effective rank, then decays to random at high d_sae. At d_sae = 1024, both match random baseline (PWMCC ~ 0.30).

4.7.3 Subspace Stability We tested whether the decoder subspace is more stable than individual features by computing principal subspace overlap across seeds.

Subspace Rank	Overlap	Random Overlap	Ratio
8	0.187	0.063	2.98x
16	0.250	0.123	2.03x
32	0.359	0.251	1.43x
64	0.552	0.497	1.11x
128	1.000	1.000	1.00x

The top-8 principal subspace is 2.98x more stable than random – substantially above the feature-level ratio of 1.02x. SAEs appear to learn a partially consistent low-rank subspace even when individual features are arbitrary, though the absolute overlap remains modest (0.187 at rank 8). This finding partially rehabilitates SAE utility: subspace-level circuit analyses and activation steering may be reliable even when feature-level interpretations are not. These subspace results were measured at the standard 20-epoch training budget. Given that individual feature stability improves substantially with extended training (Section 4.6), subspace stability may also increase; whether the 2.98x ratio at rank 8 represents a lower bound remains an open question.

4.7.4 Stability-Promoting Alignment We tested two alignment methods: a naive greedy contrastive loss and an optimal-transport (OT) alignment following Song et al. (2025). In OT alignment, we use the Sinkhorn algorithm to compute a soft permutation between decoder columns of paired SAEs, then add a cosine alignment loss on matched features with progressive warmup.

Method	PWMCC	MSE	MSE Increase
Standard (independent)	0.370	0.0102	—
Naive contrastive ($\lambda=0.1$)	0.413	0.0102	+0.0%
OT alignment ($\lambda=0.5$)	0.477	0.0102	+0.5%
OT alignment ($\lambda=1.0$)	0.567	0.0103	+1.2%
OT alignment ($\lambda=2.0$)	0.687	0.0104	+2.4%
OT alignment ($\lambda=5.0$)	0.840	0.0107	+4.9%
OT alignment ($\lambda=8.0$)	0.881	0.0108	+5.9%

OT alignment exceeds Song et al.’s (2025) 0.80 PWMCC target at $\lambda = 5.0$ with only 4.9% reconstruction cost. At $\lambda = 8.0$, PWMCC reaches 0.881 on mod-113 with a modest 5.9% reconstruction cost. The standard baseline here (0.370) reflects the 10,000-step training budget, consistent with the epoch-200 PWMCC in Section 4.6. OT alignment thus achieves in 10,000 steps what standard training has not reached even at 25,000 steps (epoch 500: 0.676), while the naive contrastive loss provides only modest improvement (+0.043).

Cross-scale validation on Pythia-410M. We replicated the OT alignment experiment on Pythia-410M layer 21 activations ($d_{\text{model}} = 1024$, $d_{\text{sae}} = 1024$, 200K cached activations, 10K steps, $\text{batch_size} = 4096$, 5 seed pairs):

Method	Mod-113 PWMCC	Pythia-410M PWMCC	Pythia MSE	Pythia MSE Δ
Standard (independent)	0.370	0.476 $\pm$ 0.016	0.341	—
Naive contrastive ($\lambda=0.1$)	0.413	0.474 $\pm$ 0.010	0.341	+0.0%
OT alignment ($\lambda=0.5$)	0.477	0.487 $\pm$ 0.017	0.342	+0.4%
OT alignment ($\lambda=1.0$)	0.567	0.495 $\pm$ 0.014	0.341	+0.1%
OT alignment ($\lambda=2.0$)	0.687	0.520 $\pm$ 0.018	0.342	+0.2%
OT alignment ($\lambda=5.0$)	0.840	0.582 $\pm$ 0.021	0.342	+0.3%
OT alignment ($\lambda=8.0$)	0.881	0.653 $\pm$ 0.031	0.347	+1.6%

OT alignment on Pythia-410M confirms that the method generalizes to LLM-scale activations: PWMCC increases monotonically with λ , reaching 0.653 at $\lambda = 8.0$ with only 1.6% MSE cost. The improvement (+0.178 over standard) is substantial and statistically robust (all 5 pairs improve). However, the absolute PWMCC (0.653) is lower than on mod-113 (0.881 at the same λ), likely because the higher-dimensional feature space (1024 vs 128) makes alignment harder. Notably, the Pythia standard baseline (0.476) already exceeds the mod-113 baseline (0.370) because these Pythia SAEs were trained for 10K steps with `batch_size` = 4096 ($\approx$ 208 epochs over 200K samples), providing more training than the mod-113 10K-step baseline ($\approx$ 157 epochs). The OT alignment improvement on Pythia (+37% relative) is smaller than on mod-113 (+138% relative), suggesting that higher-dimensional settings may require either larger λ values or more training steps to achieve the 0.80 threshold.

4.7.5 Dictionary Pinning Freezing decoder columns from a reference SAE achieves high stability but at steep cost:

Fraction Pinned	PWMCC	Reconstruction Loss
0%	0.304	21.3
6.25% (64)	0.347	20.0
25% (256)	0.478	60.3
50% (512)	0.653	123.4
75% (768)	0.828	231.1
100% (1024)	1.000	4210.4

Pinning 75% of columns exceeds Song et al.’s (2025) 0.80 target, but reconstruction degrades 11x. At low pinning fractions (6.25%), stability improves +14% with no reconstruction cost, suggesting partial pinning as a practical stabilization method.

4.7.6 Effective Rank as Diagnostic Threshold Across both 1-layer and 2-layer models, higher `d_sae / eff_rank` is consistently associated with reduced PWMCC at standard training budgets. The trend is consis-

tent but a single power-law fit achieves only $R^2 = 0.01$, indicating substantial model-specific variation. The rule of thumb is more useful than the parametric model: practitioners should measure effective rank before choosing d_{sae} .

5. Discussion

5.1 Reframing the Stability Problem

Our results reveal a more nuanced picture than either “SAEs are stable” or “SAEs match random baseline.” At standard training budgets (20 epochs, $\sim 1,000$ gradient steps), SAE features are indeed indistinguishable from random. But this reflects an undertrained regime, not a fundamental property: extended training produces 0.732 PWMCC on mod-113 (TopK, 2000 epochs, asymptotic) and 4.3-5.1x random on Pythia-scale models (200 epochs, still climbing). The convergence pattern is consistent across model scales, from 128-dim mod-113 to 1024-dim Pythia-410M.

The key insight is that the sparse reconstruction objective is underconstrained – it admits many solutions – but not all solutions are equally accessible. Given sufficient training, SAEs from different initializations converge toward similar (though not identical) decompositions. This parallels known phenomena in sparse coding for computer vision (Olshausen & Field, 1996), but adds an important temporal dimension: the landscape has many early-training basins (producing random-baseline PWMCC) that gradually funnel toward a more constrained region. Critically, this funneling effect is not task-specific: it occurs on both algorithmic (mod-113) and natural language (Pythia/wikitext) data.

A second key finding is that the $d_{\text{sae}} / \text{eff_rank}$ association with reduced stability – which was visible across 25 conditions at standard training budgets – disappears entirely with extended training. All d_{sae} values from 512 to 4096 (ratios 1.3 to 10.0) converge to 4.3-4.7x random at 200 epochs on Pythia-410M layer 21. The threshold is a training budget artifact, not a structural property.

For practitioners, this means the training budget matters critically. Standard SAE training pipelines that prioritize reconstruction loss convergence (which occurs within ~ 20 epochs) dramatically undertrain for feature stability (which requires 200+ epochs).

5.2 Implications for Interpretability

Individual feature reliability is undermined in two ways. If feature 42 appears to detect a particular concept, but a different seed assigns that concept to feature 137 (or distributes it across multiple features), the interpretation is a property of that particular run, not of the model being analyzed. Downstream circuit analyses inherit this instability: if features change with seeds, the circuits built from them change too.

The consequences extend to field-level practice. Using SAEs to verify safety properties (“no deception features detected”) is unreliable when features are unstable — a different seed might surface features the first run missed. More broadly, if each SAE run starts from a different arbitrary decomposition, studies cannot build on each other’s feature-level findings, impeding cumulative progress.

These concerns apply most directly to practitioners using standard training budgets. Our extended training results (Section 4.6) demonstrate that stability is achievable on both mod-113 and Pythia-scale models by training 10-25x longer. OT alignment (Section 4.7.4) offers a complementary path: on mod-113 it achieves

0.881 PWMCC (exceeding the 0.80 threshold) with negligible reconstruction cost, and on Pythia-410M it improves PWMCC by 37% (+0.178) with only 1.6% MSE cost. However, the diminished absolute gains on Pythia (0.653 vs 0.881 on mod-113) suggest that higher-dimensional settings require either larger alignment weights or combining OT alignment with extended training. The practical question shifts from “are SAE features stable?” to “has this SAE been trained long enough for features to stabilize?”

5.3 The Subspace Silver Lining

Our subspace stability finding (Section 4.7.3) offers a partial resolution. The top-8 decoder subspace is 2.98x more stable than random, even when individual features match random baseline. This suggests that SAEs learn a partially shared representation subspace but tile it with different basis vectors across seeds. Interpretability methods that operate on subspaces rather than individual features may be inherently more robust.

5.4 Path Forward

Immediate actions for practitioners: 1. **Train longer.** Standard training budgets (~20 epochs) dramatically undertrain for feature stability. Extended training (200+ epochs) or OT alignment yields substantially more stable features. 2. **Measure effective rank** before choosing d_{sae} . Higher $d_{\text{sae}} / \text{eff_rank}$ at short training budgets is associated with lower stability, though the strength varies by model and layer. Extended training (200+ epochs) rescues even high-overparameterization regimes. 3. **Report PWMCC alongside reconstruction metrics.** Reconstruction converges in 20 epochs; stability requires 200+. 4. **Use OT alignment** when multi-seed consistency matters. On mod-113, it exceeds the 0.80 PWMCC threshold (0.881 at $\lambda = 8.0$) with only 5.9% reconstruction cost; on Pythia-410M it reaches 0.653 (+37% over standard). 5. **Consider subspace-level analyses** when feature-level stability is insufficient.

Research priorities: 1. The TopK asymptote on mod-113 is ~0.73, below the 0.80 threshold. On Pythia, both architectures are still climbing at 200 epochs. Determine whether Pythia-scale extended training can exceed 0.80 without alignment losses. 2. Scale validation to larger models (Pythia-1B+, Llama) and longer training budgets. 3. Develop scalable stability-promoting methods (see Jedryszek & Crook, 2026 for weight regularization; Song et al., 2025 for alignment-aware training). 4. Integrate stability metrics into standard evaluation frameworks like SAEbench (Karvonen et al., 2025).

6. Limitations

1. **Scale.** Our primary experiments use a small transformer ($d_{\text{model}} = 128$). Cross-scale validation extends to Pythia-410M ($d_{\text{model}} = 1024$) but remains small by modern standards. Validation on Pythia-1B+ and frontier models is needed.
2. **Pythia training budget.** Extended training on Pythia was run for 200 epochs. On Pythia-410M layer 21, both TopK and ReLU are still climbing at epoch 200 (4.3-4.4x random). The asymptotic values on LLM-scale data are unknown and longer training may reveal further convergence or plateaus.
3. **TopK plateau on mod-113.** TopK PWMCC asymptotes at ~0.73 on mod-113 (2000 epochs), below the 0.80 consistency threshold. Whether TopK can exceed 0.80 on LLM-scale data with sufficient training is an open question.

4. **Task dependence partially mitigated.** Dense activation structure in algorithmic tasks ($\text{eff_rank} = 80/128 = 62.5\%$) may exaggerate instability. However, extended training on Pythia ($\text{eff_rank}/d_{\text{model}} = 39.9\%$) produces qualitatively similar convergence patterns, suggesting the finding is not task-specific. ReLU on Pythia-160M layer 10 reaches 5.11x random at 200 epochs, indicating real LLM representations also converge.
5. **OT alignment gains diminish at scale.** OT alignment achieves 0.881 PWMCC on mod-113 but only 0.653 on Pythia-410M at the same $\lambda = 8.0$, below the 0.80 threshold. Higher-dimensional feature spaces appear to require either larger λ values, more training steps, or architectural modifications to achieve full consistency. Whether scaling λ further or combining OT alignment with extended training can close this gap is an open question.

7. Conclusion

We provide an integrated multi-seed stability analysis combining random-baseline controls, extended training dynamics through 2,000 epochs, and cross-scale validation for SAE features. The picture that emerges is more nuanced than either “SAEs are stable” or “SAEs match random baseline.”

At standard training budgets (20 epochs, $\sim 1,000$ gradient steps), SAE features are practically indistinguishable from random (PWMCC = 0.300 vs 0.299, pairwise-comparison TOST indicates practical equivalence; 20 seeds). This result is consistent across all four architectures tested (TopK, ReLU, JumpReLU, BatchTopK) and replicates across Pythia-70M, 160M, and 410M. The effective rank diagnostic (higher $d_{\text{sae}} / \text{eff_rank}$ is associated with lower stability) holds qualitatively at standard training budgets across all 25 conditions tested, though the strength varies across models and layers.

However, extended training reveals that this instability is a property of typical training budgets, not a fundamental limitation. On mod-113, TopK SAEs asymptote at 0.732 PWMCC (2000 epochs, 2.44x random) and ReLU at 0.666. Crucially, this convergence generalizes to LLM-scale models: on Pythia-410M layer 21, both TopK (4.33x random) and ReLU (4.41x random) are still climbing at 200 epochs; on Pythia-160M layer 10, ReLU reaches 5.11x random. The $d_{\text{sae}} / \text{eff_rank}$ penalty for instability – visible at short training – disappears entirely with extended training: all d_{sae} values from 512 to 4096 converge to 4.3–4.7x random on Pythia-410M. Optimal-transport alignment achieves 0.881 PWMCC on mod-113 with only 5.9% reconstruction cost, exceeding Song et al.’s (2025) 0.80 consistency target. On Pythia-410M, OT alignment reaches 0.653 PWMCC (6.1x random, +37% over standard) with only 1.6% MSE cost, confirming the method generalizes to LLM-scale data but with diminished absolute gains in higher-dimensional settings.

The implications are practical. Standard SAE training pipelines optimize for reconstruction, which converges in ~ 20 epochs. Feature stability requires 200+ epochs or explicit alignment losses. In our Pythia-410M layer-21 sweep, the $d_{\text{sae}} / \text{eff_rank}$ penalty disappears with extended training (all dictionary sizes from 512 to 4096 converge to 4.3–4.7x random), suggesting that practitioners with sufficient training budgets need not restrict d_{sae} for stability reasons – though this result awaits replication on other models and layers. OT alignment provides a complementary path: on mod-113 it exceeds the 0.80 threshold in a single training run, and on Pythia-410M it achieves meaningful gains (+37%) with negligible reconstruction cost, though reaching the 0.80 threshold at LLM scale remains an open challenge. Until the field adopts longer training or stability-promoting objectives as standard practice, interpretability claims built on individual SAE features from standard training should be treated with caution. The tools to improve stability exist – extended training and alignment-aware objectives both show substantial gains – though achieving the 0.80 PWMCC threshold at LLM scale without alignment losses remains an open challenge.

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Appendix

A.1 Experimental Details

Full hyperparameter tables, training curves, and JSON manifest artifacts are available in the accompanying code repository. All quantitative claims in this paper trace to machine-readable experiment manifests with provenance metadata (timestamps, git commits, config hashes).

A.2 Provenance Note

This revision uses only artifact-backed numbers from experiment manifests. Claims from earlier project phases that relied on hand-entered summaries or fallback-generated values have been corrected to match verified JSON data.